# HOUSSAM EDDINE BEN KHEDER

Data Scientist & Applied Agentic AI Engineer

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## PROFILE

Data Scientist and Applied Agentic AI Engineer who builds deployed RAG and agent systems that turn large document collections into reliable, decision-ready answers. Hands-on with multimodal ingestion, hybrid retrieval, and LLM agents with planning and tool use, supported by ETL, Django and FastAPI services, and MongoDB and PostgreSQL. Takes work from prototype to production.

## EXPERIENCE

**Data Scientist & Agentic AI Engineer** — Value Digital Services
*Dec 2025 – Jun 2026 | Tunis, Tunisia*

- Led a production multimodal agentic RAG system for Tunisie Valeurs that answers analyst questions across 120+ equity research reports of 130+ pages each.
- Built the ingestion pipeline: YOLOv10 layout detection routes pages to GPT-4o Vision, so tables and charts become searchable.
- Built an agentic multimodal ETL that extracts and enriches image content into the database, with validation checks to ensure accuracy.
- Implemented a goal-driven agent with hierarchical planning, dynamic tool calling, multi-persona reasoning and long-term memory.
- Tuned hybrid retrieval (pgvector + PostgreSQL full-text) with multilingual-e5-large, so queries match both meaning and exact figures.
- Built an agentic CRM and advisory engine where deterministic KPI calculations back every conversational answer for Relationship Managers.
- Shipped the full stack (FastAPI, Next.js, Docker) with Laminar tracing for end-to-end agent observability.

**Data Science Intern** — VALUE Enterprise
*Jun 2025 – Aug 2025 | Tunis, Tunisia*

- Built agent-based simulations for macro and micro geomarketing analysis of high-value professional segments (lawyers, doctors, engineers).
- Built agent-based simulations using Monte Carlo methods and event-testing frameworks to model economic scenarios and market disruptions.
- Generated synthetic tabular data with CTGAN to address data scarcity; normalized client demographics concentrated in specific areas using a normal distribution; trained KNN models for customer segmentation.
- Added an agentic chatbot that explains simulation outputs to decision-makers and supports target-market conversion estimates.

**Machine Learning Intern** — Tritux
*Jun 2024 – Aug 2024 | Tunis, Tunisia*

- Built a telecom churn pipeline (2,666 customers, SMOTE+ENN balancing) benchmarking Logistic Regression, SVC, Gradient Boosting, XGBoost, AdaBoost and a neural network.
- Gradient Boosting reached 99.01% accuracy (AUC 1.00) on a 912-sample test set; SHAP explained churn drivers; K-Means found 3 customer clusters.
- Packaged the churn model as an MLOps workflow: tracked experiments, parameters and metrics in MLflow and registered the model in its model registry.
- Served predictions through a FastAPI service (/predict, /metrics) and a Flask app for browsing MLflow runs and artifacts; automated setup, training and serving with a Makefile.

## KEY PROJECTS

**AI-Powered Crypto Fund Due Diligence** — VALUE Enterprise & ESPRIT
*2024 – 2025*

- Built an LLM and RAG system that scores crypto fund risks across 15 dimensions (regulatory, cybersecurity, manipulation, liquidity), stored in MongoDB.
- Processed 800+ whitepaper risk questions on GPU workstations; added LSTM volatility forecasts for BTC and ETH.
- Built a PPTX reporting agent that cut due-diligence report time from 3 days to under 10 minutes.

**Deep Learning: Voice Emotion Recognition & MRI Tumor Detection** — ESPRIT
*2024*

- Emotion recognition: trained ResNet-50 embeddings on EmoDB and RAVDESS mel-spectrograms, augmented them with diffusion-generated spectrograms, and benchmarked Logistic Regression, Random Forest and MLP classifiers (Random Forest: 98.50% test accuracy, 98.26% ± 1.91% cross-validated; MLP: 97.10%).
- Brain tumour detection: evaluated a custom CNN (92% accuracy, 0.92 macro F1) against a fine-tuned VGG16 (96% accuracy, 0.96 macro F1) on a 673-sample holdout test set; VGG16 scored higher on all four classes, most on meningioma (F1 0.83 to 0.93).

## EDUCATION

**Diplôme d'Ingénieur en Informatique (Master's equivalent) — ESPRIT, Tunis**
Specialisation: Data Science & Software Engineering

**Baccalauréat en Mathématiques — Wafa School, Ariana**

## TECHNICAL SKILLS

Languages: Python, SQL, JavaScript/TypeScript
AI & Data: LLMs, Agentic AI, Multimodal RAG, Computer Vision, NLP, Time Series, Synthetic Data
Frameworks: LangChain, LlamaIndex, PyTorch, TensorFlow, Hugging Face, OpenCV, Scikit-learn, Next.js
Models: Transformers, GPT-4o, LLaMA, BERT, ResNet-50, YOLOv10, CTGAN, LSTM
Backend & MLOps: Django REST API, FastAPI, Docker, CI/CD, Git, Laminar
Data: PostgreSQL, MongoDB, pgvector, Vector Databases, Hybrid Search, ETL, Power BI
Languages: English (fluent), French (fluent), Arabic (native)